PRACTICAL NO. 7

Aim: Write a Program to implement Simple and Multiple Linear Regression Models.

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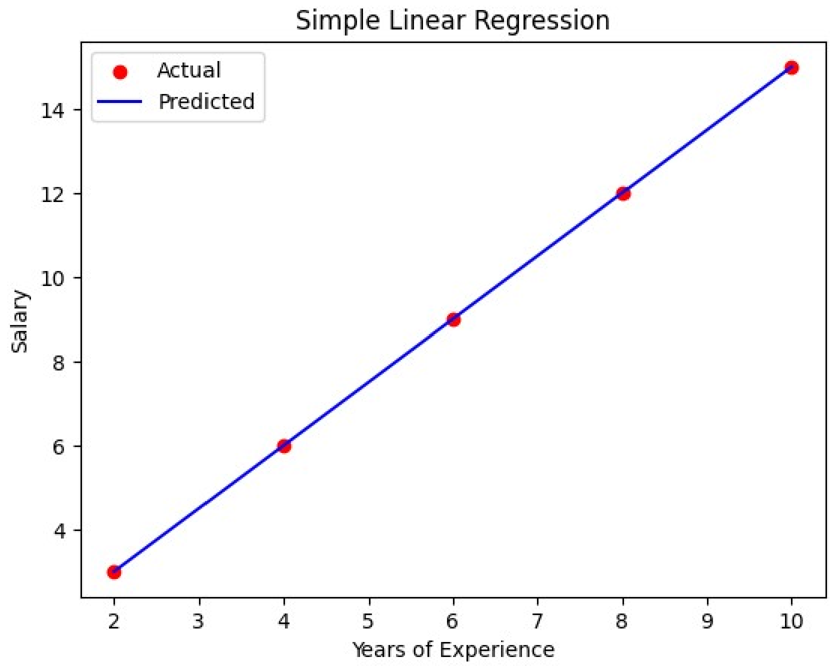
# 1) Simple Linear Regression:
import numpy as np
import matplotlib.pyplot as plt
from sklearn.linear_model import LinearRegression
# Sample dataset
X = np.array([2, 4, 6, 8, 10]).reshape(-1, 1) # Years of experience
y = np.array([3, 6, 9, 12, 15]) # Salary (in LPA, example)
# Create model
model = LinearRegression()
model.fit(X, y)
# Predict
y_pred = model.predict(X)
# Results
print("Coefficient (slope):", model.coef_[0])
print("Intercept:", model.intercept_)
# Plot
plt.scatter(X, y, color='red', label="Actual")
plt.plot(X, y_pred, color='blue', label="Predicted")
plt.xlabel("Years of Experience")
plt.ylabel("Salary")
plt.title("Simple Linear Regression")
plt.legend()
plt.show()
```

```
Coefficient (slope): 1.5000000000000004
Intercept: -3.552713678800501e-15
```




```
#2) Multiple Linear Regression
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.datasets import fetch_california_housing
california_housing = fetch_california_housing()
X = pd.DataFrame(california_housing.data, columns=california_housing.
feature_names)
y = pd.Series(california_housing.target)
X = X[['MedInc', 'AveRooms']]
X_train, X_test, y_train, y_test = train_test_split(
X, y, test_size=0.2, random_state=42)
model = LinearRegression()
```

```
model = LinearRegression()
model.fit(X_train, y_train)
y_pred = model.predict(X_test)
fig = plt.figure(figsize=(10, 7))
ax = fig.add_subplot(111, projection='3d')
ax.scatter(X_test['MedInc'], X_test['AveRooms'],
y_test, color='blue', label='Actual Data')
x1_range = np.linspace(X_test['MedInc'].min(), X_test['MedInc'].max(), 100)
x2_range = np.linspace(X_test['AveRooms'].min(), X_test['AveRooms'].max(), 100)
x1, x2 = np.meshgrid(x1_range, x2_range)
z = model.predict(np.c_[x1.ravel(), x2.ravel()]).reshape(x1.shape)
ax.plot_surface(x1, x2, z, color='red', alpha=0.5, rstride=100, cstride=100)
ax.set_xlabel('Median Income')
ax.set_ylabel('Average Rooms')
ax.set_zlabel('House Price')
ax.set_title('Multiple Linear Regression Best Fit Line (3D)')
plt.show()
```

```
/usr/local/lib/python3.12/dist-packages/sklearn/utils/validation.py:2739: UserWarning: X does not have valid fea
  warnings.warn(
```